
Exploration of Rewards Shaping and Imitation Learning in Chess

This project explores the interaction between **Imitation Learning** and **Reward Shaping** in training chess agents. We evaluate how pretraining on **puzzle positions and grandmaster games** affects subsequent **REINFORCE policy gradient** fine-tuning across **FC**, **CNN**, and **ResNet** architectures.

Report

Our report can be found under Report.

Presentation

A presentation as a video can be found under Presentation with the corresponding Slides.

Installation

For installation and setup instructions, see the Quickstart Guide.

Features

- imitation learning on master games and puzzle positions
- REINFORCE policy gradient fine-tuning
- Support for multiple neural architectures: FC, CNN, ResNet
- Evaluation across different combinations of pretraining and reward sets